

Problem Statement – Smart Curriculum Activity & Attendance App

Domain: Smart Education

Problem ID: IH-01

Background

Many educational institutions continue to rely on manual attendance systems, which consume valuable class time and are prone to errors. Additionally, students often spend free periods unproductively due to the lack of structured academic guidance or personalized planning. Institutions also lack a centralized platform that combines attendance management with daily academic planning and student engagement.

Problem Description

Develop a **Smart Curriculum Activity & Attendance Web Application** that automates classroom attendance using **QR code-based verification** while helping students utilize their free periods effectively.

The application should provide separate interfaces for **students** and **teachers**. Teachers should be able to generate attendance sessions, view attendance records, and monitor class participation. Students should be able to mark attendance by scanning a QR code, view their attendance percentage, access their daily timetable, and receive personalized academic task suggestions during free periods.

The system should also generate a simple daily planner based on the student's class schedule and identify available free periods to encourage productive learning. Additionally, an analytics dashboard should provide attendance insights for both students and faculty.

The solution should be lightweight, user-friendly, and capable of being demonstrated as a functional prototype within the hackathon duration.

Expected Solution

The proposed solution should include the following core features:

Teacher Module

- Secure teacher login.
- Create and manage classes.
- Generate QR code for attendance.
- View real-time attendance status.
- Access attendance history and reports.

Student Module

- Secure student login.
- Scan QR code to mark attendance.
- View attendance percentage.
- Access daily timetable.
- View personalized daily planner.
- Receive suggested academic activities during free periods.

Analytics Module

- Attendance statistics.
 - Daily attendance trends.
 - Student attendance percentage.
 - Class-wise attendance summary.
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Core Features (Mandatory)

- Student Login
 - Teacher Login
 - QR Code-Based Attendance
 - Attendance Record Management
 - Daily Timetable Display
 - Free Period Detection
 - Personalized Daily Planner
 - Attendance Analytics Dashboard
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Bonus Features (Optional)

Teams may implement any of the following for additional consideration:

- Export Attendance Reports (PDF/Excel)
 - Admin Dashboard
 - Dark Mode
 - And features you prefer additionally
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Expected Deliverables

Each team should submit:

- Functional software prototype
- Source code
- Database schema
- UI/UX design
- Live demonstration (5–7 minutes)
- Teams must maintain and submit the complete source code in a **public GitHub repository**.
- The developed prototype should be **deployed on a cloud platform** such as Render, Netlify, Vercel, or any suitable hosting service, with a working deployment link.